Bayesian Optimisation fo Discrete Event Simulation Experiments

Harry Newton

2026-04-08

1 Experiment: Pilot_01 VIHGP

In this experiment we compare the performance between three configurations:

- *Vanilla*: VI-HGP GP with Expected Improvement
- *AEI*: VI-HGP with Augmented Expected Improvement
- *IG* VI-HGP with Information Gain

All three experiments began with k equally spaced intial values. Each was sampled n times to produce a dataset $\mathcal{D} = (x_i, y_i)_{i=1}^{k \times n}$. Each data point can be summarised by the set (x, n, y). Each experiment was run over M macroreplications and each BO instance was run over T iterations. If Vanilla configuration has been included in these experiments it has no replication aware acquisition function, so n_v replications are sampled every iteration. The table bellow shows the values used for this experiment:

Table 1: Experimental paramaters.

Paramater	Value
k	50
M	25
T	10
n	1
n_v	5
seed	12345

1.1 Test Problem

The test function for this experiment is defined as:

$$f(x) = \sin(5x) + \cos(7x).$$

And the noise function is defined as:

$$\sigma_\varepsilon^2(x) = (0.3 + \tau \exp(-0.5((x-\phi)/0.1)^2))^2$$

The paramatarisation of the noise function, the x and n domains can be found in the table below.

Table 2: Test Problem paramater values and domain. Replication domain is also included

Paramater	Value
x	$[0, 1]$
n	$[1, 100]$
ϕ	0.6
τ	0.7

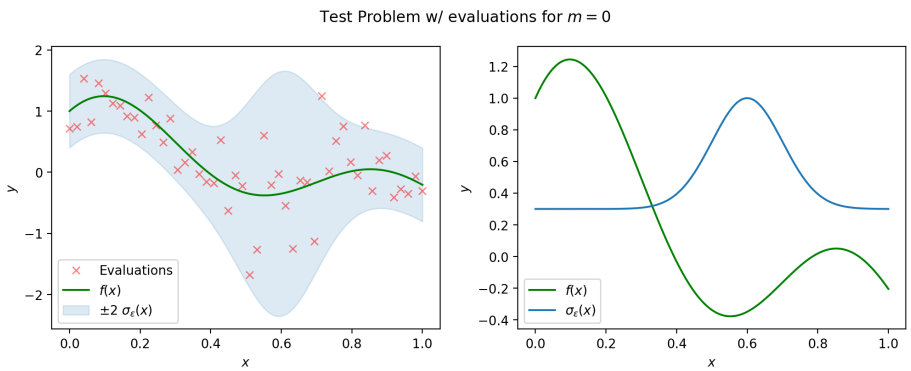

Figure 1: Test problem to be optimised with the true signal and noise functions present. Initial evaluations for $m = 22$ are also present

The cost function is the inverse linear function

$$c(n) = \frac{1}{\beta_0 + \beta_1 n}$$

Table 3: Cost function paramaters.

Paramater	Value
β_0	1
β_1	0.5

1.2 Variational Paramaters

Table 4: This table includes the number of iterations used to fit the VI-HGP and the number of inducing points used - n_u.

Paramater	Value
iters	800
n_u	10

1.3 Results

In the results section below I compare the performance between the three configurations: vanilla, IG and AEI.

A basic indicator of performance is the effectiveness of a configuration in finding the optimal point with minimal expenditure of resources. Here we plot the average cumulative sum of $\hat{\mathbf{n}}$, which is the average replication selection path over $M = 25$ macroreplications for each iteration, against the average true maximum f^*- the best recorded f^* up to a given BO iteration t. The $\hat{\mathbf{x}}^* = [x_1^*, \dots, x_T^*]$ sequence is the GP model's predicted maximiser of the test function f for each iteration t, then the predicted optimal solutions can by found by inputting these values into the test function $\hat{\mathbf{f}}^* = f(\hat{\mathbf{x}}^*)$.

In the section below are two plots (1) The average true best solution against cumulative number of replications. Also included in this plot are the 10th (dashed line) as well as the 90th (dotted line) percentiles, and the true optimal value f^*.(2) The average number of replications spent per iteration.

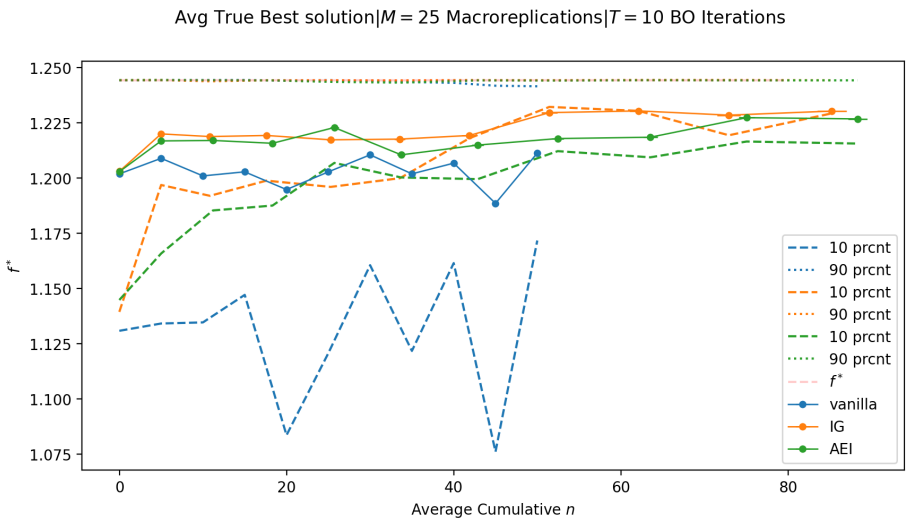

Figure 2: Performance Comparison between models across 25 macroreplications using the maximum predicted solution. This is calculated using the predicted maximiser and the true, noiseless test function.

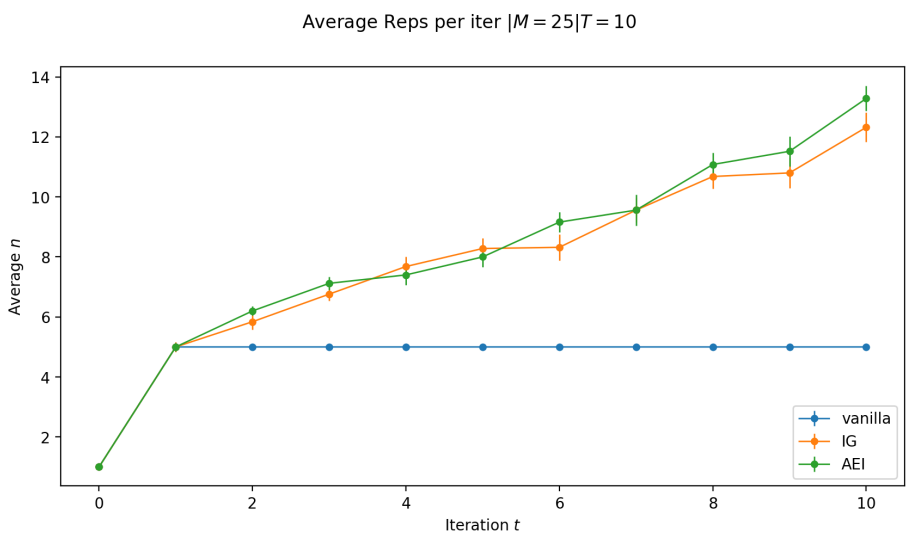

Figure 3: Average 'number of replications' n selected pre iteration. Compared over models